Batching and epochs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Name the batch. Update the current state. Count the full pass.

Trace schedule-dependent updates and aggregate losses with correct case weights.

Keep one habit: Declared epoch rule → complete two-pair traces → pictured order counterexample → weighted frozen errors → independent new order comparison.

MAIN LESSON · SECTION 1

When does the model change during a pass through the data?

> Count examples,batches,updates and epochs separately.

Learning goals

I can distinguish an example a **batch** an **update** and an **epoch**.

I can trace parameter changes across successive **batch**es.

I can calculate **update** counts and aggregate unequal **batch** losses correctly.

Retrieve the complete training loop

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Week 12 printed states numbered 0 to 4. How many parameter **update**s occurred?

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

End goal

By the end, I can explain **batch**ing and **epoch**s using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Distinguish an example a **batch** an **update** and an **epoch**

Trace parameter changes across successive **batch**es

Calculate **update** counts and aggregate unequal **batch** losses correctly

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Declare one update per nonempty batch and keep the last batch

> An epoch visits each training row once under this stated sampling rule.

Complete batch/update/epoch convention

In Week 12 every **update** used both **training** examples. We now keep the same network h = ReLU(x), y = vh, with only v trained, and divide the data into **batch**es. In this lesson each nonempty **batch** produces exactly one **update**.

An **epoch** visits every **training** example exactly once, without replacement. These rules make the counts unambiguous.

Imagine five fictional records A through E in fixed order. With **batch size** 2, the groups are [A, B], [C, D] and [E]. Keep the smaller last **batch**.

One **epoch** has three **batch**es and three **update**s; two **epoch**s have six **update**s and ten example visits.

Five-record schedule counts

Training rows	Batch size	Batch lengths	Updates per epoch
5	2	2, 2, 1	3
5	5	5	1

Training rows	Batch size	Batch lengths	Updates per epoch
5	1	1, 1, 1, 1, 1	5

Retained versus dropped incomplete batch

When the last **batch** is retained, **update**s per **epoch** equal the row count divided by **batch size** rounded up. A program that drops the final incomplete **batch** follows a different rule and omits examples in that pass.

Record the rule instead of silently changing the denominator.

Name supplied and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Trace a parameter state before and after every batch

> A full-batch mean and successive single-example updates are different schedules.

Complete data and batch-gradient definition

Use the two fictional pairs A = (1, 2) and B = (2, 4), start v = 0 and rate 0.1. For one pair with hidden value h, squared-error gradient is 2 times (vh - target) times h.

For a **batch**, average these gradients at the same current v. Apply one **update**, then recompute for the next **batch**.

Complete current-state batch traces

Main data A(1, 2), B(2, 4); start v0, rate.1	Current v before batch	Gradient at this v	New v after batch
Size1:Afirst	0	-4	0.4
Size1:Bnext	0.4	-12.8	1.68
Restart size 1:Bfirst	0	-16	1.6
Then A	1.6	-0.8	1.68
Restart size 2:AandBtogether	0	Mean(−4, −16)=−10	1

Two-record versus five-ID fixture and helper contract

The main executable uses exactly the two supplied numeric training pairs, not the five-ID counting example. Each new batch recomputes its gradient at the latest v; within one batch all per-row gradients use the same vbefore its one update. The main A/Borders happen to share final v1.68 but have different paths.

That coincidence does not prove order never matters; the pictured example uses different targets and shows different final weights. The helper requires nonempty rows and size≥1, and assumes integer size and finite numeric pairs/rate.

Assume finite numeric pairs, integer size and finite rate. With size 1, A gives gradient -4 and new v = 0.4. B then uses 0.4, giving **prediction** 0.8, residual -3.2 and gradient -12.8, so v becomes 1.68.

With size 2, both gradients are evaluated at v = 0; their mean is -10, producing v = 1.

Runnable retained-last-batch epoch

```
def epoch(v, rows, size, rate):
    if not rows or size < 1:
        raise ValueError("Need rows and positive batch size")
    trace = []
```

```
    for start in range(0, len(rows), size):
        batch = rows[start:start + size]
        hs = [max(0.0, x) for x, target in batch]
        gradient = sum(2 * (v * h - target) * h
                       for h, (x, target) in zip(hs, batch)) / len(batch)
        v = v - rate * gradient
        trace.append(v)
    return trace

rows = [(1.0, 2.0), (2.0, 4.0)]
print([round(v, 6) for v in epoch(0.0, rows, 1, 0.1)])  # [0.4, 1.68]
print([round(v, 6) for v in epoch(0.0, rows, 2, 0.1)])  # [1.0]
```

Connect the diagram to update timing

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change the batching schedule while retaining its full record

> An update happens after a batch,not automatically after every example.

Current stale-gradient task

- For A then B separately in the lesson, a learner computes both gradients at v = 0, then applies them to get v = 2. Explain the error and give the correct final value.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the pictured reversed full batch

- Change: Use only one full-**batch update** after reversing record order.

▼ **Check after predicting**

Prediction: The **update**d weight remains 0.2.

Reason: The same mean of the two starting gradients is unchanged by their order.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

See why order can matter after single-example updates

> The next gradient sees the changed weight;the full-batch mean sees one saved state.

Complete pictured data and distinct scope

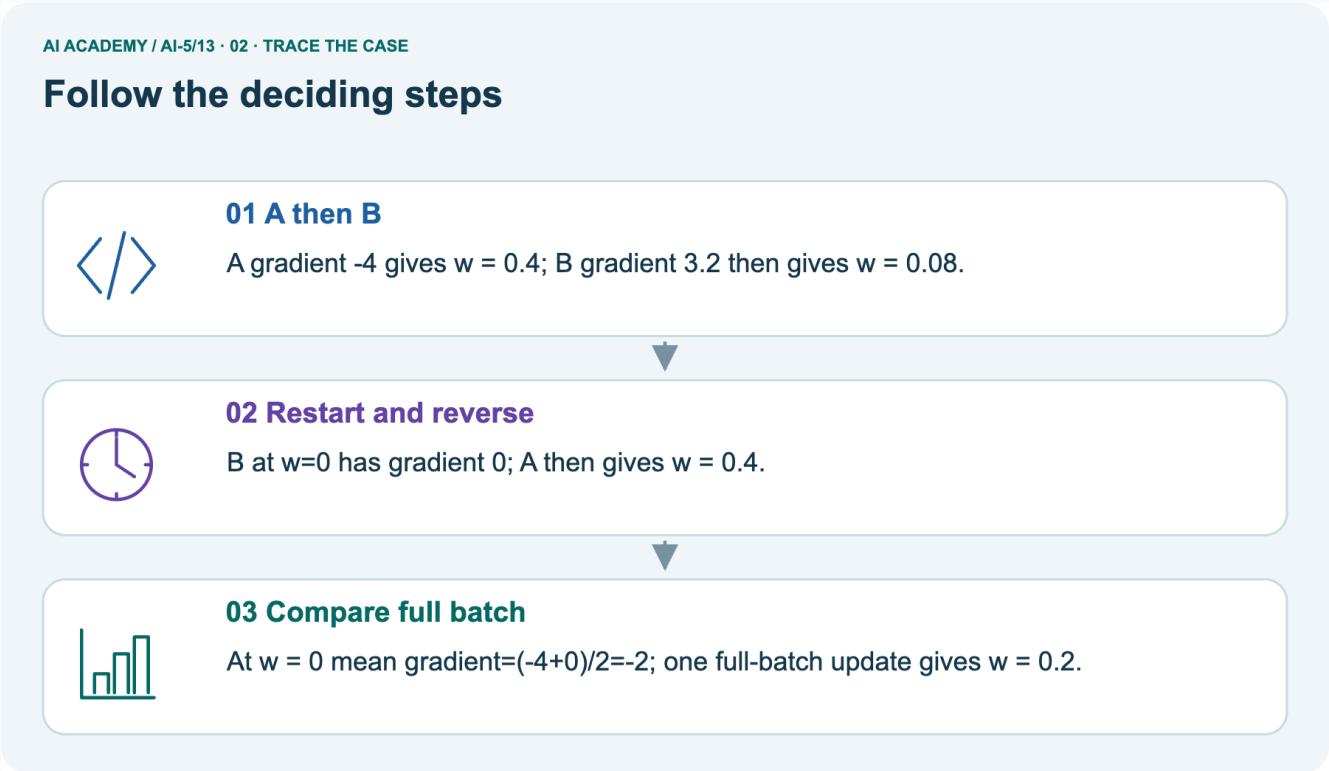
Show why an **epoch**'s example order can matter for single-example **update**s.

Toy y=wx, rate 0.1, squared loss, start w = 0. **Training** A=(x = 1, target 2), B=(x = 2, target 0).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses y=wx, start w0, rate.1, A(1, 2), B(2, 0), squared loss. AthenB:size 1 gradients−4 then3.2, give w.4 then.08. Restart BthenA:gradients 0 then−4, give w0 then.4. Full-batch mean at the same old w0is(−4+0)/2=−2, giving w.2; reversing the rows leaves that exact mean unchanged. These targets differ from the main A(1, 2)/B(2, 4)and the independent A(1, 1)/B(2, 0).

Trace both orders and the full-batch mean



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. A then B	A gradient -4 gives w = 0.4; B gradient 3.2 then gives w = 0.08.
2. Restart and reverse	B at w=0 has gradient 0; A then gives w = 0.4.
3. Compare full batch	At w = 0 mean gradient=(-4+0)/2=-2; one full-batch update gives w = 0.2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

Predict one reversed full-batch update

- Change one thing: Use only one full-**batch update** after reversing record order. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-5/13 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

At w = 0 mean gradient=(-4+0)/2=-2; one full-batch update gives w = 0.2.

Changed condition

Use only one full-batch update after reversing record order.

Worked changed-case result

The updated weight remains 0.2.
Why: The same mean of the two starting gradients is unchanged by their order.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain update timing and fair-comparison limits

- Different order effects do not mean a universally better **batch**ing strategy; compare declared schedules and budgets fairly.

- Read the labelled picture: Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main reversed-order trace

One pass can contain different numbers of updates

BATCH RULE	PARAMETER AFTER EACH BATCH	UPDATES
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

For B then A separately, show both gradients and new parameter values. Why does one epoch still contain two updates?

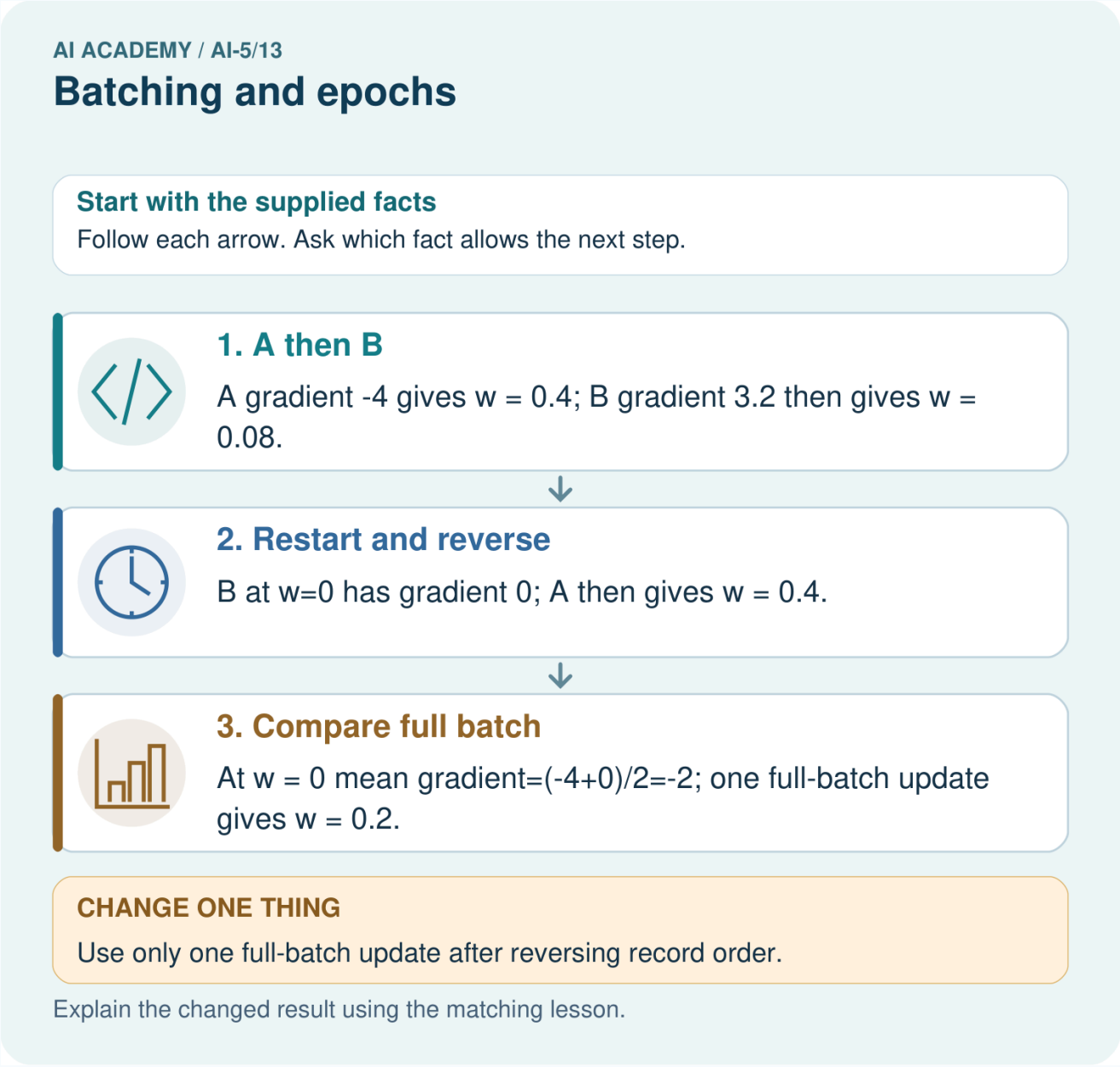
▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Batching and epochs

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Freeze one model before aggregating its loss

> Weight unequal batch means by the number of cases they represent.

Training-time state changes and weighted aggregation

Batch losses observed during **training** can refer to different parameter values and different examples. They are not automatically the full **training** loss of the final model.

To compare final models, freeze each one and evaluate every **training** example again. Here full **training** MSE is 2.5 times (v - 2) squared: at v = 1 it is 2.5, and at v = 1.68 it is 0.256. These are **training** results only.

Even at one fixed model, averaging **batch** means without their sizes gives the wrong dataset mean when **batch**es have different lengths. Suppose the squared errors on A to E are [1, 1, 4, 4, 9].

Batch means for lengths 2, 2, 1 are [1, 4, 9]. Full MSE is (2 times 1 + 2 times 4 + 1 times 9) / 5 = 3.8, not (1 + 4 + 9) / 3. Use example counts as weights.

Complete five-error weighted example

Separate frozen-model squared-error fixture	Batch size	Batch MSE	Contribution to total squared error
A/Berrors1/1	2	1	2
C/Derrors4/4	2	4	8
Eerror9	1	9	9
All five cases	5	(2+8+9)/5=3.8	19
Wrong unweighted mean of means	Not one five-case average	(1+4+9)/3=14/3	Overweights the single Ecase

Try sizes3and1with different means

- At one frozen model, **batch**es of sizes 3 and 1 have mean squared errors 2 and 10. Find the full MSE and explain why the average of 2 and 10 is wrong.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise schedule counts and state timing with less help

> Keep update counts and example visits beside every reported epoch.

Current task and response

- For 10 rows, **batch size** 4 and three **epoch**s, retaining the final **batch**, list **batch** lengths and count total **update**s and example visits.

For B then A separately, show both gradients and new parameter values. Why does one epoch still contain two updates?

> **Explain your prediction to a partner before comparing results.**

Supported W2record

Use workbook W2 for the connected example “**Batch**ing and **epoch**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Batch**ing and **epoch**s. Proposed explanation: “One **epoch** always means one **update**”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured order reasoning

- Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the independent new data

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Repair a stale gradient

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete schedule,state and aggregation audit

> Each fixture declares whether it supplies numeric training pairs,IDs for counting or frozen errors.

Current stale-gradient task and runnable epoch

For A then B separately in the lesson, a learner computes both gradients at v = 0, then applies them to get v = 2. Explain the error and give the correct final value.

```
def epoch(v, rows, size, rate):
    if not rows or size < 1:
        raise ValueError("Need rows and positive batch size")
    trace = []
    for start in range(0, len(rows), size):
```

```
        batch = rows[start:start + size]
        hs = [max(0.0, x) for x, target in batch]
        gradient = sum(2 * (v * h - target) * h
                       for h, (x, target) in zip(hs, batch)) / len(batch)
        v = v - rate * gradient
        trace.append(v)
    return trace

rows = [(1.0, 2.0), (2.0, 4.0)]
print([round(v, 6) for v in epoch(0.0, rows, 1, 0.1)])  # [0.4, 1.68]
print([round(v, 6) for v in epoch(0.0, rows, 2, 0.1)])  # [1.0]
```

Complete schedule,invalid,missing and weighted-loss probes

Complete batching probe	Expected result / action
Normal two rows, size 1, AthenB	Two updates:v0→.4→1.68
Same two rows, size 2	One mean-gradient update:v0→1
Five ID-only rows, size 2, retain last	Batch lengths 2/2/1; 3 updates per epoch; 2 epochs=6 updates/10 visits
Five rows, size 5, 2 epochs	2 updates/10 visits; same visits, different update budget
Empty rows or size 0	ValueError:Need rows and positive batch size
Missing numeric training target	Outside complete-pair contract; review before calling
Eleven rows, size 4, drop incomplete last	Only8 visitsand2 updates; disclose 3 omitted rows, not full coverage
Frozen batch sizes 3/1, mean losses 2/10	Full MSE(3×2+1×10)/4=4, not unweighted 6
Both gradients computed at originalv 0 then applied sequentially	Stale Bgradient−16 givesv2; correct current Bgradient−12.8 givesv1.68

Predict,restart,trace or run,and aggregate correctly

1. Predict the complete schedule counts and numeric traces before running or paper-tracing epoch on its two provided numeric pairs. The five-ID schedules illustrate counts only; the separate five-error fixture illustrates frozen-model loss aggregation. Do not invent training measurements for those IDs.

   For each numeric schedule, restart v0, record each batch's current state and recomputed gradient, and evaluate the full training loss again after freezing. A passing count or lower training loss does not establish a universally better batch strategy or independent performance.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: reverse a new pair’s single-example schedule

> Use its own targets and restart the same original value before reversing.

Complete new independent schedule

Use A = (1, 1), B = (2, 0), h = ReLU(x), y = vh, v = 0 and rate 0.1. Train with size 1 for one **epoch** in A then B order, then restart at 0 and reverse the order. Show gradients and parameter values. Compare the final values.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Test order on a new case independently

Attempt this without hints

Use A = (1, 1), B = (2, 0), h = ReLU(x), y = vh, v = 0 and rate 0.1. Train with size 1 for one **epoch** in A then B order, then restart at 0 and reverse the order. Show gradients and parameter values. Compare the final values.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the schedule as part of the experiment

> Example visits,updates,order and last-batch policy all matter to reproducibility.

Project artifact and evidence table

Add a labelled **batch**ing and **epoch**s **evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish an example a **batch** an **update** and an **epoch**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and remaining limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one update count and one state-dependent gradient

> Explain why equal epochs need not mean equal numbers of updates.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Distinguish an example a **batch** an **update** and an **epoch**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Batch	A group of training examples used together to calculate one gradient here
Update	One application of the gradient and learning rate to the current parameters
Epoch	One pass through all training examples under the stated sampling rule
Batch size	The intended number of examples in each batch with a possible smaller final batch

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For 10 rows, **batch size** 4 and three **epoch**s, retaining the final **batch**, list **batch** lengths and count total **update**s and example visits.
- At one frozen model, **batch**es of sizes 3 and 1 have mean squared errors 2 and 10. Find the full MSE and explain why the average of 2 and 10 is wrong.
- A program **process**es 11 rows using size 4 and drops the last incomplete **batch**. How many examples did it visit and how many **update**s occurred? What must its report disclose?

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent A(1,1)/B(2,0)order comparison for the first individual attempt.

Current explanation and response

- Equal **epoch**s mean equal example visits under our rule, but changing **batch size** changes the number of **update**s. A comparison should report both counts, starting values, order, learning rate and the rule for the last **batch**.

  Neither smaller **batch**es nor more **update**s guarantee better independent performance.

- Next week holds the **batch** rule fixed while comparing learning rates.

- A program **process**es 11 rows using size 4 and drops the last incomplete **batch**. How many examples did it visit and how many **update**s occurred? What must its report disclose?

Two teams train on five rows for two epochs. One uses size 2 and the other size 5, retaining every row. Are their example visits and update counts equal? Explain what to record.

My limited claim and a useful next check

▼ Extra practice · Revisit the complete trace

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

One pass can contain different numbers of updates

BATCH RULE	PARAMETER AFTER EACH BATCH	UPDATES
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Show why an **epoch**'s example order can matter for single-example **update**s.

By the end, I can	How I will show it
Distinguish an example a batch an update and an epoch	Explain the deciding step, show a trace or test, and state one limit.
Trace parameter changes across successive batches	Explain the deciding step, show a trace or test, and state one limit.
Calculate update counts and aggregate unequal batch losses correctly	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Show why an **epoch**'s example order can matter for single-example **update**s.
- Which labelled step matches this detail? At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2.
- 1. Restart and reverse
- 2. A then B
- 3. Compare full **batch**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare full **batch** At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2.

A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

Changed-case question and optional hint

- Changed-case practice: Use only one full-**batch update** after reversing record order. Which conclusion is supported by the supplied facts?
- 1. The **update**d weight remains 0.2.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The **update**d weight remains 0.2. The same mean of the two starting gradients is unchanged by their order.

Repair and retain the record

Review **Training** a small network (ai-5/12). Goal: Trace repeated forward loss gradient and **update** steps. Short practice: Which parameter changed, and which part was not trained?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

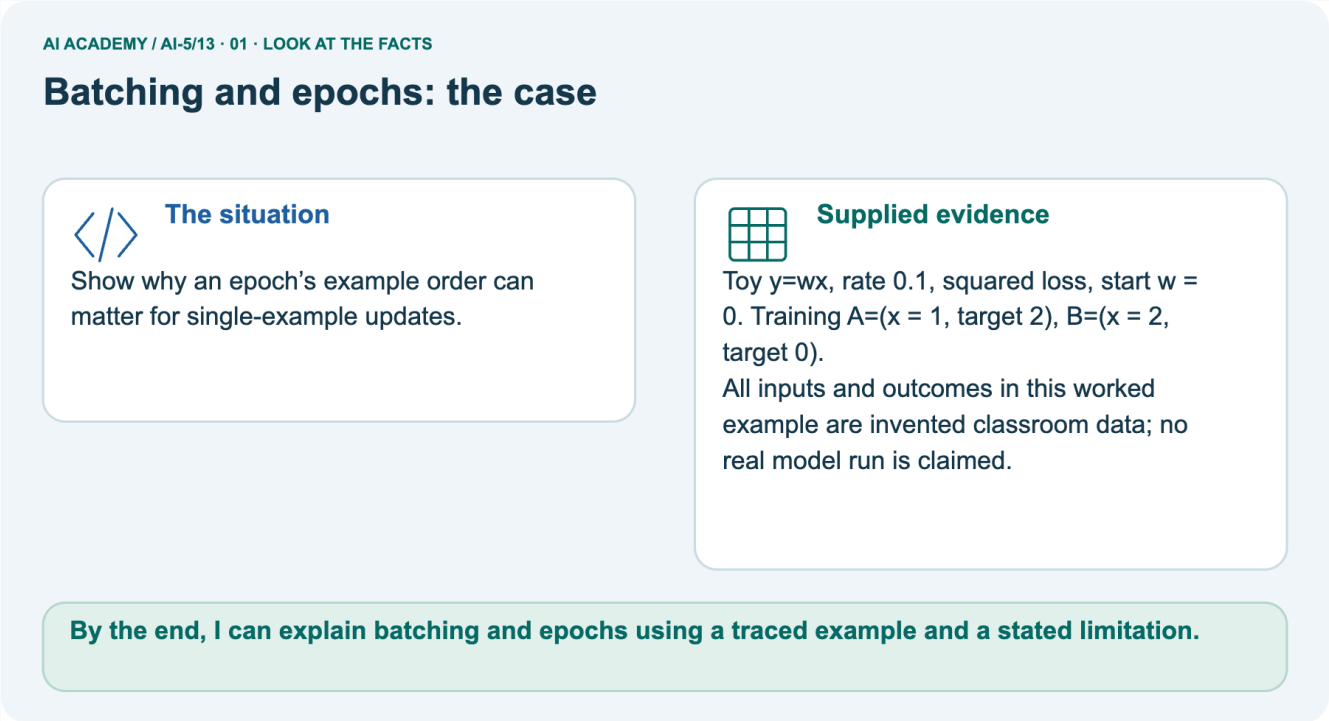
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Batching and epochs: the case



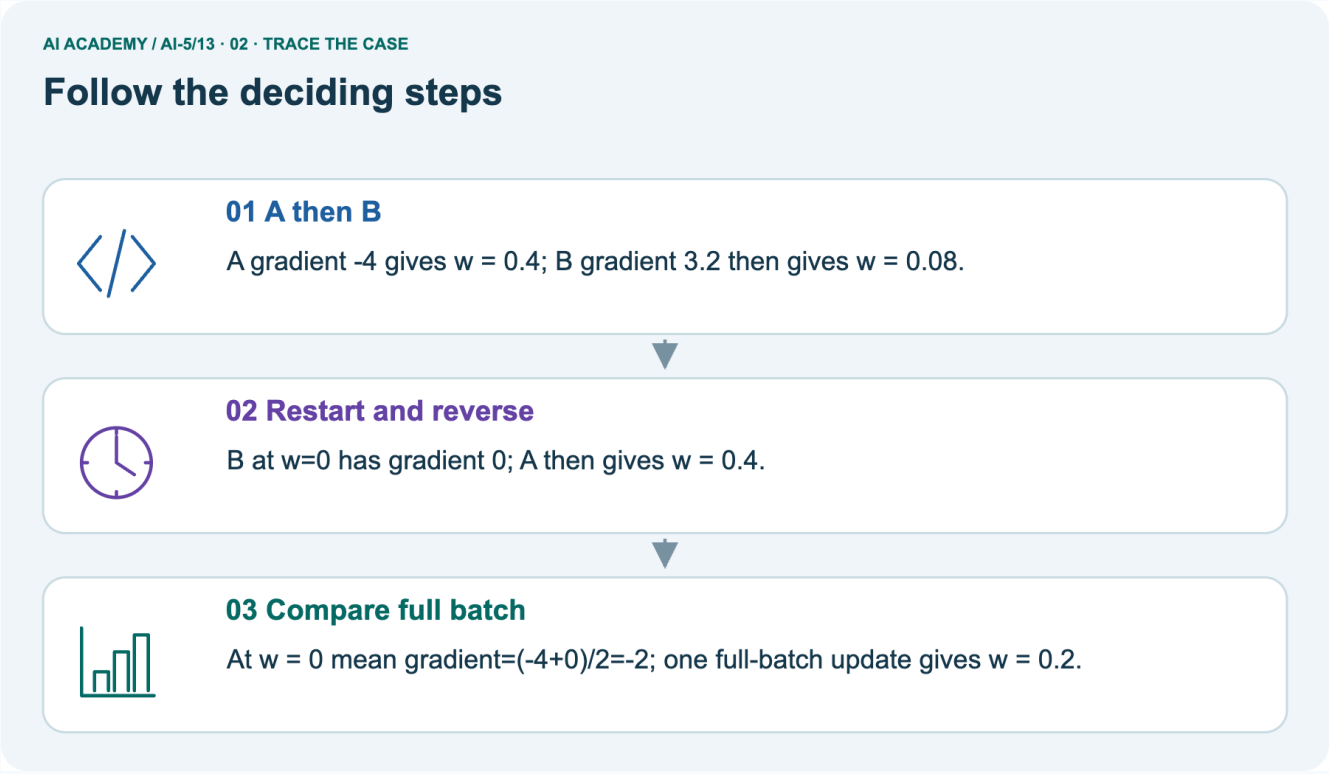
Follow the labelled picture with the matching explanation. Open the vector diagram

Show why an **epoch**'s example order can matter for single-example **update**s. Supplied facts: Toy y=wx, rate 0.1, squared loss, start w = 0. **Training** A=(x = 1, target 2), B=(x = 2, target 0).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

A then B: A gradient -4 gives w = 0.4; B gradient 3.2 then gives w = 0.08. → Restart and reverse: B at w=0 has gradient 0; A then gives w = 0.4. → Compare full **batch**: At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2.

Point to each numbered stage and explain which recorded **input** it uses.

03. One pass can contain different numbers of updates

AI ACADEMY / AI-5/13 · 03 · INSPECT THE LESSON MODEL

One pass can contain different numbers of updates

Batch rule	Parameter after each batch	Updates
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

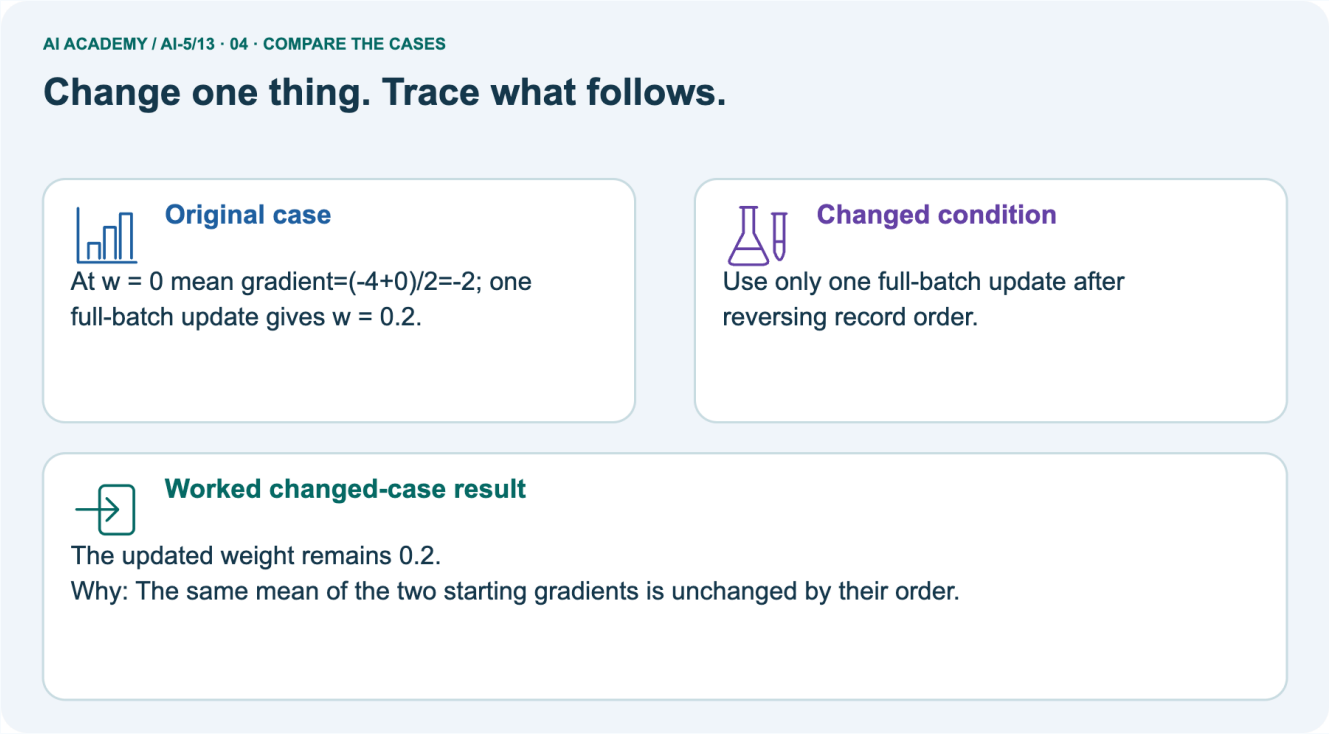
All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

Follow the labelled picture with the matching explanation. Open the vector diagram

One pass can contain different numbers of **update**s. A then B separately | 0 to 0.4 to 1.68 | 2. A and B together | 0 to 1 | 1. B then A separately | 0 to 1.6 to 1.68 | 2

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



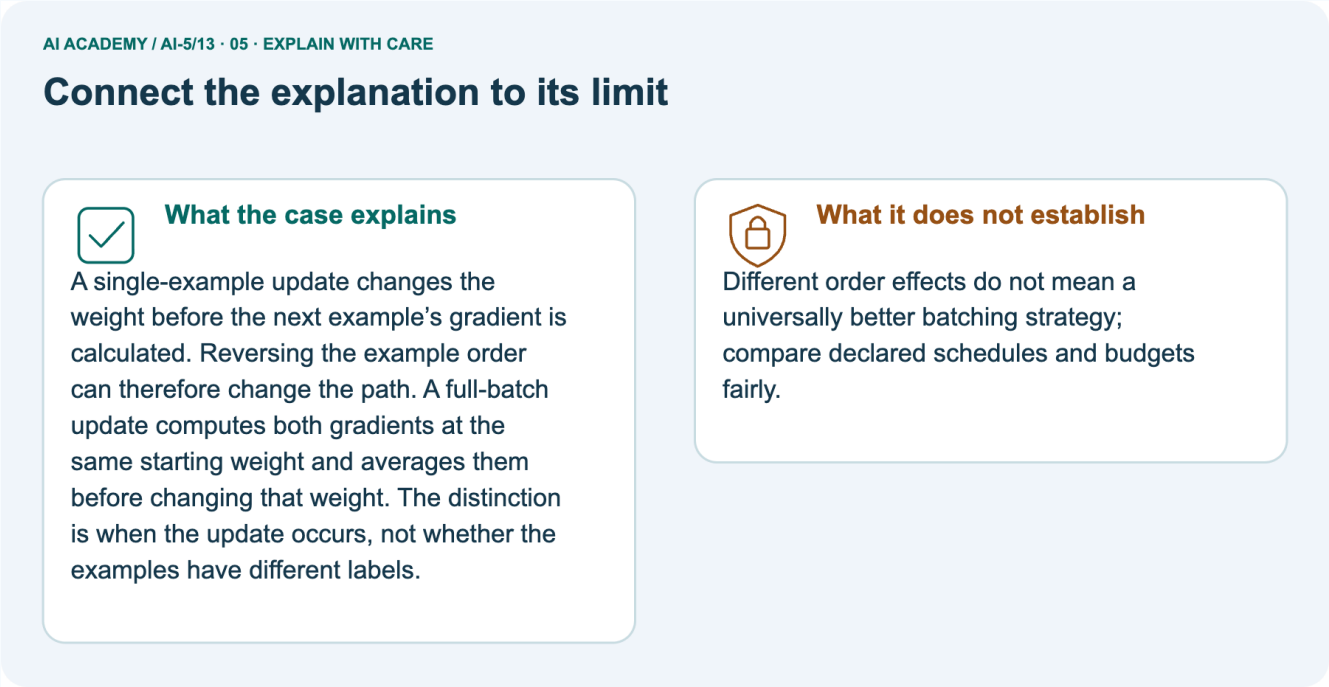
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2. Changed condition: Use only one full-**batch update** after reversing record order. Expected result: The **update**d weight remains 0.2.

Reason: The same mean of the two starting gradients is unchanged by their order.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

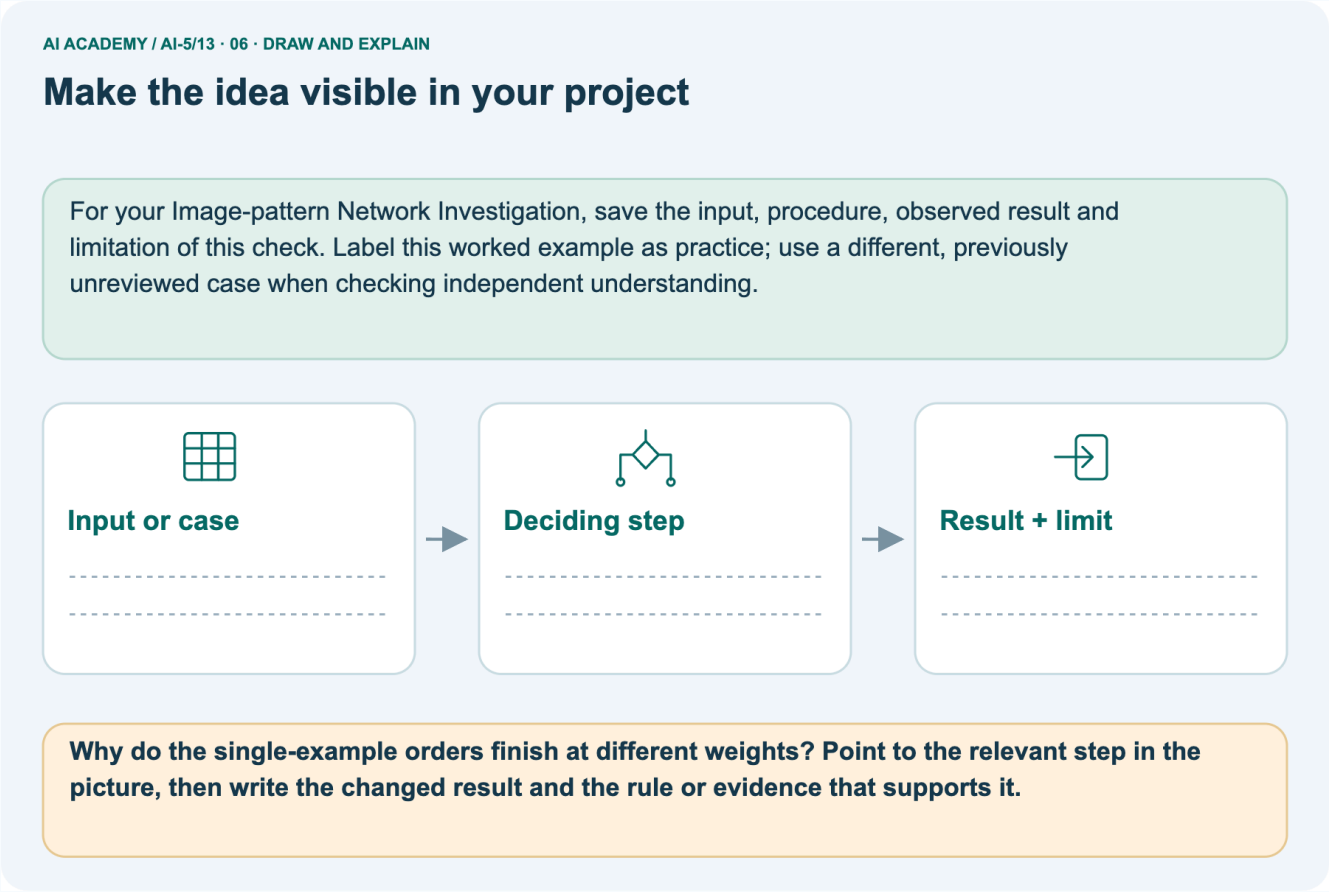
A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

Limit: Different order effects do not mean a universally better **batch**ing strategy; compare declared schedules and budgets fairly.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Batching and epochs: the case

02. Follow the deciding steps

03. One pass can contain different numbers of updates

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.